

Electrochemistry

Igcse Chemistry

Electrolysis

Electrolysis 电解 is the breaking down (**decomposition** 分解) of an **ionic compound** 离子化合物—when it is **molten** 熔融 or in **aqueous** 水溶液 solution —by passing an **electric current** 电流 through it.

It only works when the substance is molten or dissolved, because then the **ions** 离子 are free to move and carry the charge. A solid ionic compound cannot be electrolysed because its ions are locked in place.

The electrolytic cell

The set-up is called an **electrolytic cell** 电解池. Two **electrodes** 电极 (solid conductors) dip into the **electrolyte** 电解质—the molten or aqueous substance being broken down.

- The **anode** 阳极 is the positive (+) electrode.
- The **cathode** 阴极 is the negative (−) electrode.

The electrodes are often **inert** 惰性 (they do not react), such as **platinum** 铂 or carbon/**graphite** 石墨.

What is formed at each electrode

There is a simple rule:

- **Metals** 金属 or **hydrogen** 氢气 are formed at the cathode.
- **Non-metals** 非金属 (other than hydrogen) are formed at the anode.

For aqueous solutions, the product can depend on whether the solution is **dilute** 稀 or **concentrated** 浓. Here are the three Core examples:

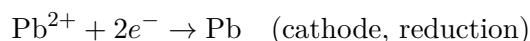
Electrolyte	At the cathode (−)	At the anode (+)
molten lead(II) bromide, PbBr_2	lead 铅 (silvery liquid)	bromine 溴 (red-brown vapour)
concentrated aqueous sodium chloride	hydrogen (bubbles of gas)	chlorine 氯气 (pale green gas)
dilute sulfuric acid 硫酸	hydrogen (bubbles of gas)	oxygen 氧气 (bubbles of gas)

How the charge moves

During electrolysis:

- **Electrons** 电子 move through the wires (the **external circuit** 外电路) from the power supply.
- At the cathode, positive ions gain electrons. Gaining electrons is **reduction** 还原.
- At the anode, negative ions lose electrons. Losing electrons is **oxidation** 氧化.
- Inside the electrolyte, the ions move: positive ions go to the cathode and negative ions go to the anode.

You can write a **half-equation** 半反应式 for each electrode. For molten lead(II) bromide:



Electrolysis of copper(II) sulfate

The product at the anode depends on the electrode:

- With **inert** carbon electrodes: **copper** 铜 forms at the cathode and oxygen forms at the anode. The blue colour of the solution slowly fades as copper is removed.
- With copper electrodes: copper forms at the cathode, while the anode itself dissolves into the solution. The blue colour stays the same. This is used to purify copper.

Electroplating

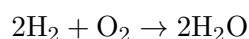
Electroplating 电镀 means covering a metal object with a thin layer of another metal. This improves its appearance and its resistance to **corrosion** 腐蚀 (rusting and wearing away).

To electroplate an object:

- the object to be coated is made the **cathode**;
- the plating metal is made the **anode**;
- the electrolyte is a solution containing ions of the plating metal.

Hydrogen–oxygen fuel cells

A **fuel cell** 燃料电池 uses hydrogen and oxygen to make electricity directly. The only chemical product is water.



It is useful to compare a fuel cell with a normal petrol (gasoline) engine in a vehicle.

	Hydrogen–oxygen fuel cell	Petrol engine
Main product	water only	carbon dioxide and pollutants 污染物
Effect on air	clean	adds to air pollution
Fuel storage	hydrogen is hard and dangerous to store (flammable, needs high pressure)	petrol is easy to store
Source	hydrogen may be made using fossil fuels	made from crude oil

Advantages of the fuel cell: the only product is water, so it does not pollute the air, and it changes chemical energy into electricity efficiently. **Disadvantages**: hydrogen is hard to store and transport safely, and producing the hydrogen can still use energy from fossil fuels.